How Routing Works in ASP.NET Core Web API

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Routing is a component of ASP.NET Core framework that **associates application** URL with a controller action. Every Web Page, or Service API endpoint in ASP.NET Core the process is passed by the **Routing Middleware**. Routing also uses the URL to generate outgoing URL that can be used to invoke a specific action.

Before understanding how Routing Works in ASP.NET Core Web API, let's first understand the core components where the routing is implemented.

Understanding Core Routing Components in ASP.NET Core

Routing is implemented in ASP.NET Core using **Middleware**, **Endpoints**, and **Route Tables**. Let's quickly take a look at each of these components in ASP.NET Core routing system. For a more detailed explanation of these components, see the ASP.NET Core routing system in ASP.NET Core Web API, or in the underlined the core components where the routing is implemented.

Routing Middleware

- **RouteMiddleware**
- **EndpointRoutingMiddleware**
- **EndpointMiddlewareBase**

Together, they form the backbone of the **Endpoint Routing** system. We expect the controller routing office in the middle of the middleware pipeline, ensuring that the middleware can dynamically manage routing, which allows for flexible and efficient request processing and high performance.

Endpoints

At the heart of Endpoint Routing System is the **Endpoint** class. Think of it as the **Universal Contract** that represents a single, executable route in the application. Each endpoint encapsulates a unique URL pattern, a set of HTTP methods, and a request delegate. This key idea is that the routing middleware doesn't care what it's routing to. It only cares that it's routing to an endpoint.

Route Tables & Collections

ASP.NET Core manages these endpoints in two primary ways:

- **RouteEndpoint**: A powerful, feature-rich endpoint that remembers when this endpoint is matched. Its capability that enable complex routing scenarios.
- **Endpoint**: A simpler, lightweight, and generic endpoint. Its primary use is for endpoints, such as diagnostics, health checks, or other simple, non-routing services.

The **EndpointDataSource** is a key abstraction that provides a collection of endpoints. Routing, in Endpoint Routing, then queries this data source to find the matching endpoint.

Endpoint Metadata & Other Components

The extensible metadata system is a key aspect of modern routing. This metadata can be used for things like authorization, CORS, and more. Some other important components are representation metadata, **Endpoints**.

Other Routing Components

- **Routing Middleware**: Manages the overall routing process.
- **Endpoint Selector**: The component responsible for selecting the appropriate endpoint from the matched candidates.
- **Endpoint Metadata**: Provides additional information about an endpoint, such as authorization requirements, CORS policies, or custom data.
- **Route Constraints**: Enforce additional rules on route parameters, such as data types or regular expressions.
- **Route Parameters**: Extract dynamic segments from the URL and make them available to your application.

Route Resolution Process

The routing system matches incoming requests against the defined endpoints. This process involves parsing the URL, extracting route parameters, and evaluating any applicable constraints. The framework performs a series of checks to ensure a match. This ensures that only valid requests are routed to the correct endpoint.

Extensibility

The routing system is designed for extensibility. You can create custom route constraints, custom endpoint metadata, and even custom endpoint data sources to tailor the routing behavior to your specific needs. This flexibility allows you to handle complex routing scenarios and integrate with other parts of your application.

Endpoint Routing

Endpoint Routing is a powerful technique used in ASP.NET Core to decouple routing from the actual execution of endpoints. It allows for more flexible and efficient routing, especially in complex applications. The key components involved in Endpoint Routing are:

- **Endpoint**: Represents a single, executable route in the application.
- **EndpointDataSource**: Provides a collection of endpoints.
- **Route Tables**: The routing system maintains internal tables that map incoming requests to endpoints.
- **Middleware Integration**: The routing middleware integrates with other middleware in the pipeline.

Endpoint Routing offers several advantages:

- **Performance**: It can be more efficient than traditional routing, especially for complex applications.
- **Flexibility**: It allows for more complex routing scenarios.
- **Extensibility**: The system can be easily extended with custom endpoints.
- **Integration**: It integrates well with other ASP.NET Core features like authorization and CORS.

Understanding Endpoint Routing is crucial for optimizing performance and leveraging the full power of ASP.NET Core's routing system. It's a foundational concept that underlies many advanced routing scenarios in ASP.NET Core Web API.

How Routing Works in ASP.NET Core Web API

In the ASP.NET Core Web API, routing is the process of matching incoming HTTP requests to the appropriate controller and action. This process is crucial for directing requests to the correct part of your application. Here's how routing works in ASP.NET Core Web API:

Step 1: Application Startup

When an ASP.NET Core application starts, the **Program.cs** file is first executed. In this file, the application configuration is set up, including the routing system. The routing system is initialized and configured to work with the Web API.

```
public class Program
{
    public static void Main(string[] args)
    {
        var builder = WebApplication.CreateBuilder(args);
        // Add services to the container.
        builder.Services.AddControllers();
        // Learn more about configuring Swagger/OpenAPI at https://aka.ms/aspnetcore/swashbuckle
        builder.Services.AddEndpointsApiExplorer();
        builder.Services.AddSwaggerGen();
        var app = builder.Build();
        // Configure the HTTP request pipeline.
        if (app.Environment.IsDevelopment())
        {
            app.UseSwagger();
            app.UseSwaggerUI();
        }
        app.UseHttpsRedirection();
        app.UseAuthorization();
        app.MapControllers();
        app.Run();
    }
}
```

Why does the application work even though there's no explicit app.UseRouting()?

Starting from .NET 6, ASP.NET Core introduced the **Minimal Hosting Model**, where routing is automatically configured by default. When you call:

```
var app = builder.Build();
app.MapControllers();
```

The framework **implicitly** adds the **Routing Middleware** and **Endpoint Middleware**. If you prefer, you can explicitly add them, but it's not required. For more details, see the official documentation.

Step 2: Middleware Configuration

At this stage, the **Routing Middleware** is added to the pipeline. This step is key, as it allows the application to process incoming requests and determine which endpoint should handle each request.

Step 3: Route Patterns Are Set Up

Controllers and their action methods are discovered, and their route patterns are established. These patterns define the URL structures that will be matched against incoming requests.

Step 4: Incoming Requests Are Matched

When a request comes in, the routing system matches the request URL against the defined route patterns. This matching process considers HTTP methods, URL segments, and any constraints defined.

Step 5: Controller and Action Are Selected

Once a match is found, the routing system selects the appropriate controller and action method to handle the request. This selection process considers factors such as HTTP method, URL parameters, and any custom routing rules.

Step 6: Request Pipeline Is Executed

The request pipeline is then executed, including any additional middleware that may be configured. This pipeline allows for cross-cutting concerns like authentication, logging, and error handling.

Step 7: Response Is Generated

The selected action method is executed, and a response is generated. This response can be a JSON object, HTML, or any other format supported by your application.

Step 8: Response Is Sent to the Client

The response is then sent back to the client through the HTTP protocol.

Step 9: Routing Middleware Processes Request

When a request comes in, the Routing Middleware processes it. This middleware is responsible for matching the request URL against the defined route patterns.

Step 10: Endpoint Selection

The routing system selects the appropriate endpoint based on the matched route. This selection process considers factors such as HTTP method, URL parameters, and any custom routing rules.

Step 11: Middleware Execution

The selected endpoint's middleware is then executed. This includes any authorization, CORS, or other middleware that has been configured for this specific endpoint.

Step 12: Controller Action Is Invoked

If the endpoint is associated with a controller, the framework creates an instance of the controller and invokes the appropriate action method.

Step 13: Response Is Generated

The action method processes the request and generates a response. This response can be a JSON object, HTML, or any other format supported by your application.

Step 14: Response Is Sent to the Client

The response is then sent back through the middleware pipeline and finally to the client through the HTTP protocol.

Step 15: Logging and Monitoring

Throughout this process, various components may log information about the routing process, which can be useful for debugging and monitoring the application's performance.

Step 16: Error Handling

If any errors occur during routing, the framework will attempt to handle them gracefully, potentially returning a 404 Not Found response if no matching route is found.

Step 17: Performance Optimization

The routing system is optimized for performance, using techniques like caching route information and compiled expressions to speed up the matching process.

Step 18: Extensibility

Developers can extend the routing system by adding custom route constraints, custom endpoint metadata, and even custom endpoint data sources to tailor the routing behavior to specific needs.

Step 19: Integration with Other Features

The routing system integrates with other features of ASP.NET Core, such as authorization, CORS, and API versioning, to provide a comprehensive solution for handling HTTP requests.

Step 20: Continuous Improvement

The routing system is continually improved with each new version of ASP.NET Core, with Microsoft adding new features and optimizations based on feedback and performance analysis.

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Step 2: Middleware Configuration

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Step 3: Routing Processes the Service Container

The routing system processes the service container to find the **Endpoint Sources** (IEs) provided. These sources are responsible for defining the available endpoints in the application. Without an routing, HTTP Request, a set of endpoints, and a matcher, the routing system cannot function.

Step 4: Endpoint Sources Are Set Up

The **Endpoint Sources** are then set up to provide the **Endpoints**. These endpoints are the actual routes that the routing system will use to match incoming requests. The Endpoint Sources can include controllers, Razor Pages, minimal APIs, SignalR hubs, and more.

Step 5: Incoming Requests Are Matched

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